Synchronous Machines



LEARNING OBJECTIVES

- Understand the equivalent circuit of a synchronous motor.
- Be able to sketch phasor diagrams for a synchronous motor.
- Know the equations for power and torque in a synchronous motor.
- Understand how and why power factor varies as synchronous motor load increases.
- Understand how and why power factor varies as synchronous motor field current varies—the "V" curve.
- Understand how synchronous motors can be started.
- Be able to tell whether a synchronous machine is acting as a motor or a generator and whether it is supplying or consuming reactive power by examining its phasor diagram.
- Understand synchronous motor ratings.

Construction of Synchronous Machines

Synchronous machines are AC machines that have a field circuit supplied by an external DC source.

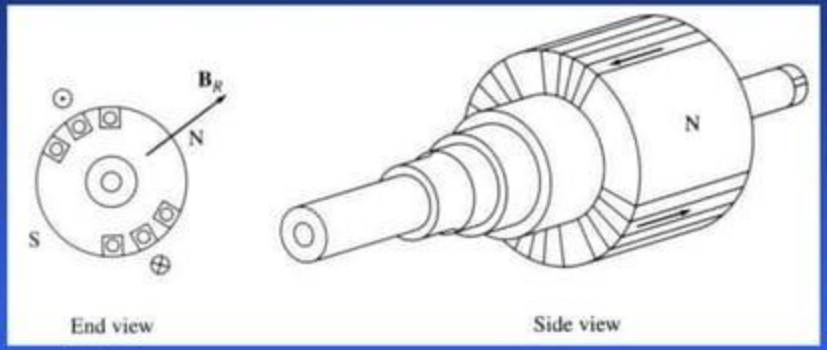
In a synchronous generator, a DC current is applied to the rotor winding producing a rotor magnetic field. The rotor is then turned by external means producing a rotating magnetic field, which induces a 3-phase voltage within the stator winding.

In a synchronous motor, a 3-phase set of stator currents produces a rotating magnetic field causing the rotor magnetic field to align with it. The rotor magnetic field is produced by a DC current applied to the rotor winding.

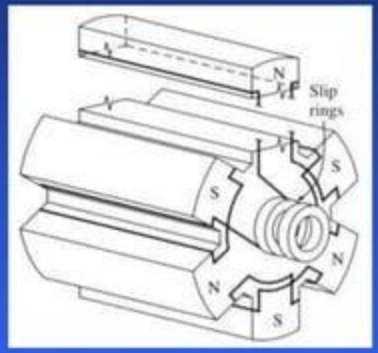
Field windings are the windings producing the main magnetic field (rotor windings for synchronous machines); armature windings are the windings where the main voltage is induced (stator windings for synchronous machines).

Construction of Synchronous Machines

The rotor of a synchronous machine is a large electromagnet. The magnetic poles can be either salient (sticking out of rotor surface) or non-salient construction.



Non-salient-pole rotor: usually two- and four-pole rotors.

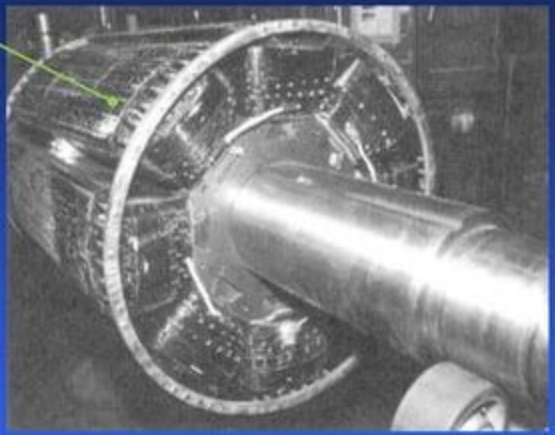


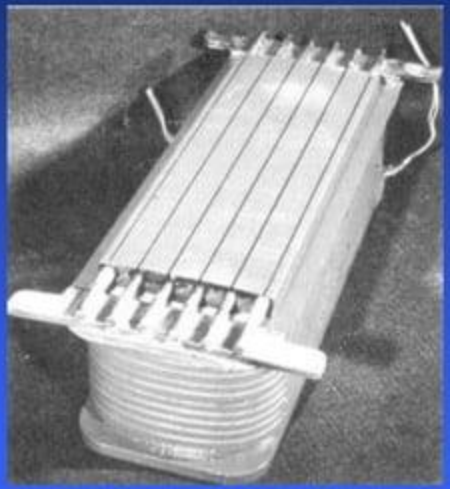
Salient-pole rotor: four and more poles.

Rotors are made laminated to reduce eddy current losses.

Construction of Synchronous Machines

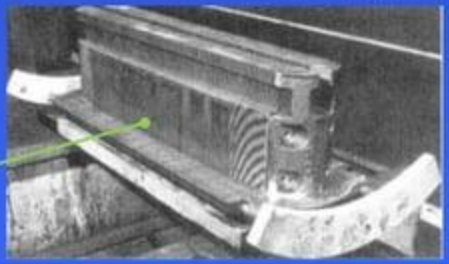
A synchronous rotor with 8 salient poles





Salient pole with field windings

Salient pole without field windings – observe laminations



Construction of Synchronous Machi

Two common approaches are used to supply a DC current to the field circ
the rotating rotor:

1. Supply the DC power from an external DC source to the rotor by means of slip rings and brushes;

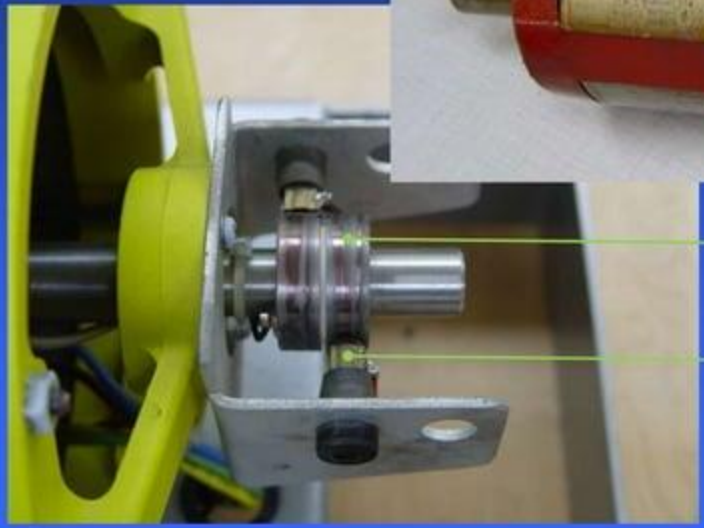
2. Supply the DC power from a special DC power source mounted directly on the shaft of the machine.



Slip rings are metal rings completely encircling the shaft of a machine b
from it. One end of a DC rotor winding is connected to each of the two s
the machine's shaft. Graphite-like carbon brushes connected to DC termi

Construction of Synchronous Machines





Construction of Synchronous Machines

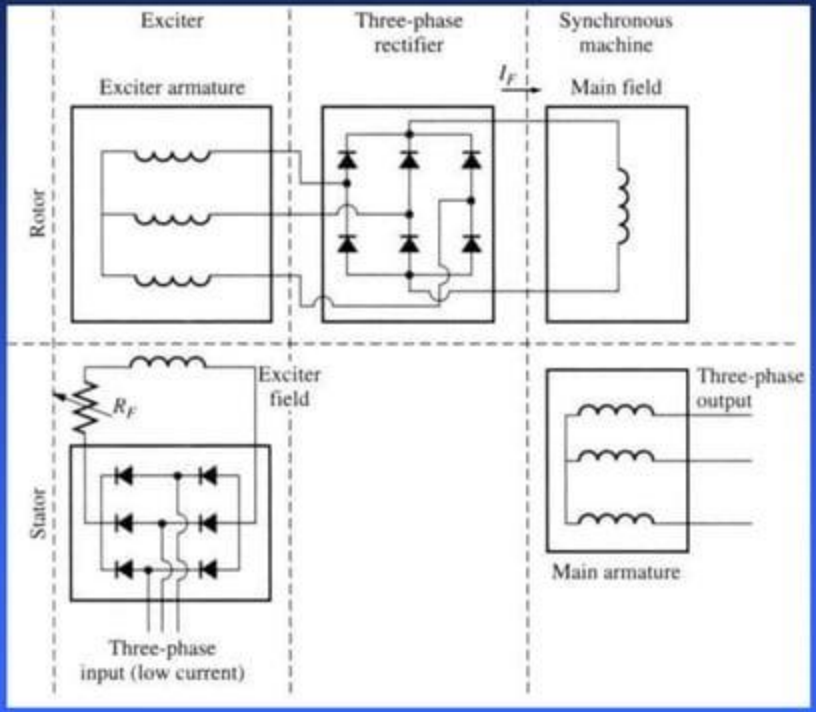
Slip rings and brushes have certain disadvantages: increased friction and wear (therefore, needed maintenance), brush voltage drop can introduce significant power losses. Still this approach is used in most *small* synchronous machines.

On large generators and motors, brushless exciters are used.
A brushless exciter is a small AC generator whose field circuits are mounted on the stator and armature circuits are mounted on the rotor shaft. The exciter generator's 3-phase output is rectified to DC by a 3-phase rectifier (mounted on the shaft) and fed into the main DC field circuit. It is possible to adjust the field current on the main machine by controlling the small DC field current of the exciter generator (located on the stator).

Since no mechanical contact occurs between the rotor and the stator, exciters of this type require much less maintenance.

Construction of Synchronous Machines

A brushless exciter: a low 3-phase current is rectified and used to supply the field circuit of the exciter (located on the stator). The output of the exciter's armature circuit (on the rotor) is rectified and used as the field current of the main machine.



Starting Generator (Previous slide)

1. **Start-up phase:**
Initially, the excitation power is provided by an **external DC or AC source** (e.g., a battery, auxiliary power supply, or the grid) to create the initial magnetic field in the exciter and build up the generator voltage.

2. Normal operation (self-excitation):
Once the generator output voltage has been established, a portion of the **main generator's own output** is fed back (via transformers or potential transformers) to power the exciter stator winding. This is known as **self-excited operation**.

3.Voltage regulation:

The **Automatic Voltage Regulator (AVR)** continuously monitors the generator output voltage. By adjusting the current supplied to the exciter stator (via controlled rectification or a controlled power source), the AVR regulates the main field strength and thereby maintains a constant generator terminal voltage under varying loads.

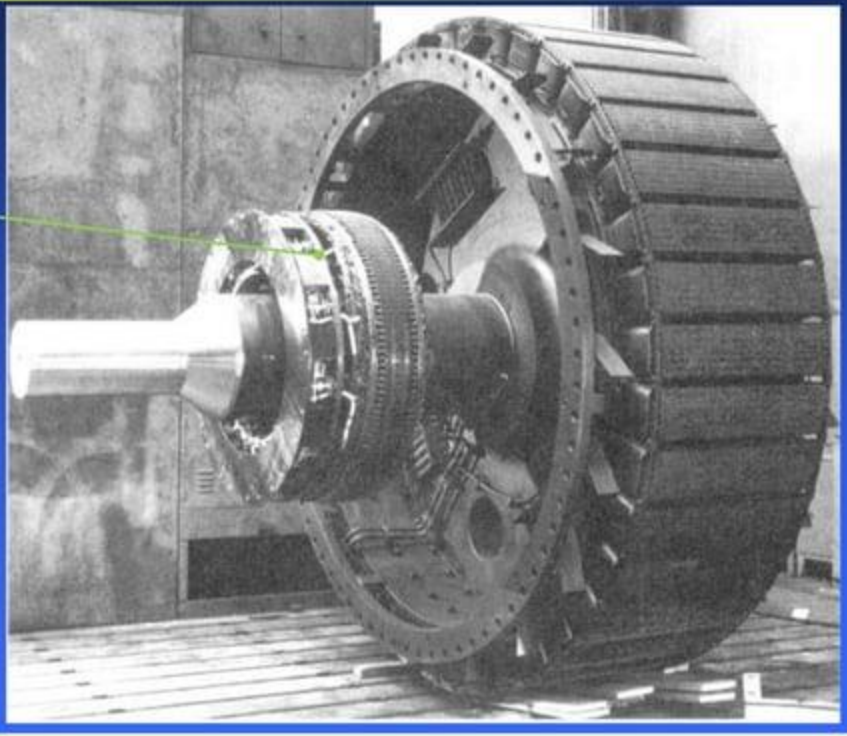
In practice,

1. First, it comes from an external source (to start up the system).
3.Then it is supplied from the main generator's own output (self-excited mode).
4.The AVR continuously adjusts this supply to keep the main output voltage constant.

Construction of Synchronous Machines

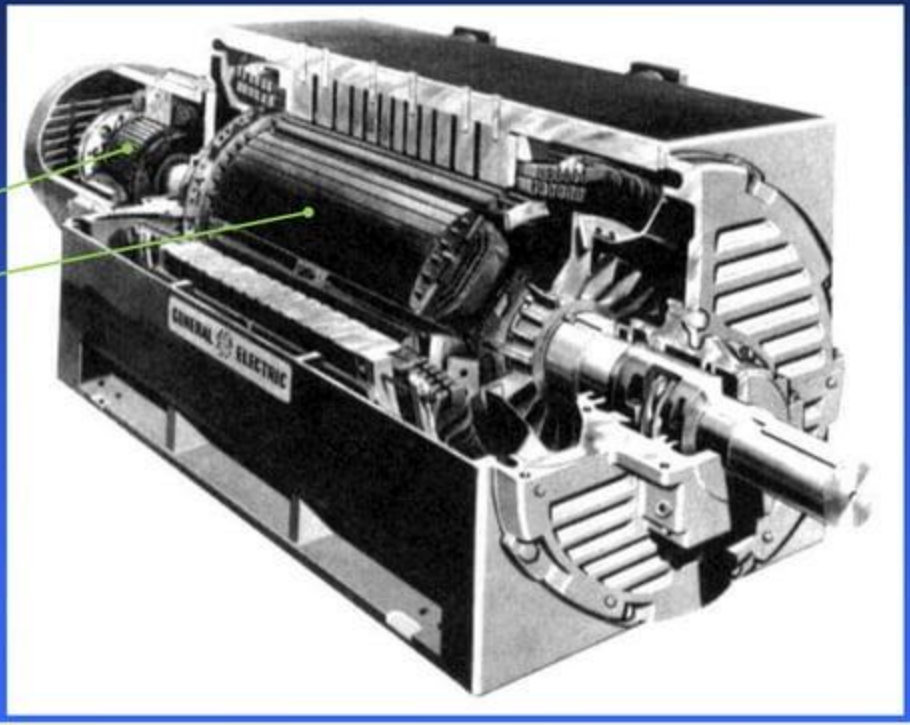
A rotor of large synchronous machine with a brushless exciter mounted on the same shaft.

Many synchronous generators having brushless exciters also include slip rings and brushes to provide emergency source of the field DC current.



Construction of Synchronous Machines

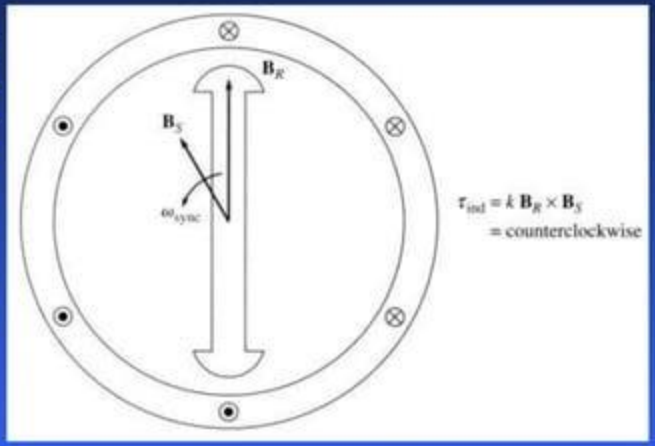
A large synchronous machine with the exciter and salient poles.



Synchronous motors

The field current I_F of the motor produces a steady-state rotor magnetic field B_R. A 3-phase set of voltages applied to the stator produces a 3-phase current flow in the windings.

A 3-phase set of currents in an armature winding produces a uniform rotating magnetic field B_S.



Two magnetic fields are present in the machine, and the rotor field tends to align with the stator magnetic field. Since the stator magnetic field is rotating, the rotor magnetic field will try to catch up pulling the rotor.

The larger the angle between two magnetic fields (up to a certain maximum), the greater the torque on the rotor of the machine.

OPERATION PRINCIPLE

Under no load:

Rotor magnetic field locks perfectly with the stator's rotating magnetic field. They rotate in sync (slip = 0).

Under load:

1. The rotor poles are "pulled back" slightly relative to the stator field due to load torque.
2. This constitutes a load angle (δ) between the rotor field axis and the stator field axis.
3. The rotor never perfectly aligns with the instantaneous position of the stator field — it chases it, lagging by the load angle.

This is a steady-state phase lag, not a speed difference.

OPERATION PRINCIPLE

- The rotor still rotates at exactly synchronous speed, but its magnetic axis is always a few degrees behind the stator's magnetic axis under motor operation.
- If the load torque increases too much, the rotor loses synchronism (pull-out).

Starting Methods for Synchronous Motors

- Synchronous motors cannot self-start (due to rotor inertia and hunting effects), so various starting methods are employed.
- Here are the most common starting methods for synchronous motors:

1. Asynchronous (Induction) Starting (Most Common)

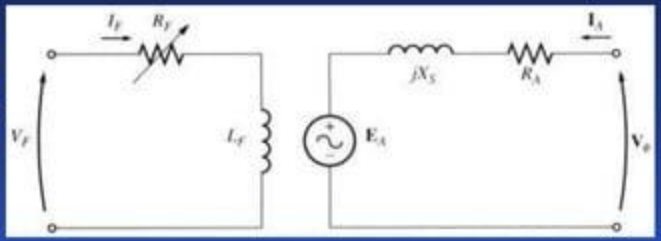
- Damper windings (damper bars/squirrel cage) are mounted on the rotor pole surfaces.
- The motor initially starts like an induction motor (with slip).
- When it approaches synchronous speed (slip ≈ 2–5%), DC field excitation is applied.
- The rotor locks into the stator's rotating magnetic field.
- Then the damper winding carries no significant current (becomes inactive due to the synchronization).
- DC excitation is not interrupted; it remains on continuously.

2. Inverter / Soft Starting (Using VFD)

- A variable frequency voltage is applied to the stator.
- Frequency is gradually increased from zero.
- The rotor field (DC excitation) is energized from the beginning.
- The rotor follows the slowly increasing stator field and stays synchronized.

Synchronous motor equivalent circuit

A synchronous motor has the same equivalent circuit as synchronous generator, except that the direction of power flow (and the direction of I_A) is reversed. Per-phase circuit is shown:



A change in direction of I_A changes the Kirchhoff's voltage law equation:

$$V_\phi = E_A + jX_S I_A + R_A I_A \qquad (7.69.1)$$

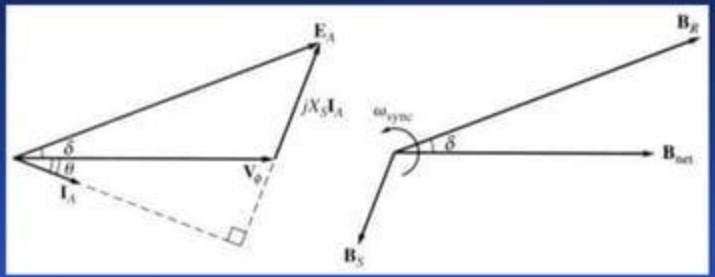
Therefore, the internal generated voltage is

$$E_A = V_\phi - jX_S I_A - R_A I_A \qquad (7.69.2)$$

We observe that this is exactly the same equation as the equation for the generator, except that the sign on the current terms is reversed.

Synchronous motor vs. synchronous generator

Let us suppose that a phasor diagram of synchronous generator is shown. B_R produces E_A, B_{net} produces V_ϕ and B_S produces $E_{stat} = -jX_S I_A$. The rotation on both diagrams is counterclockwise and the induced torque is



$$\tau_{ind} = kB_R \times B_{net} \qquad (7.70.1)$$

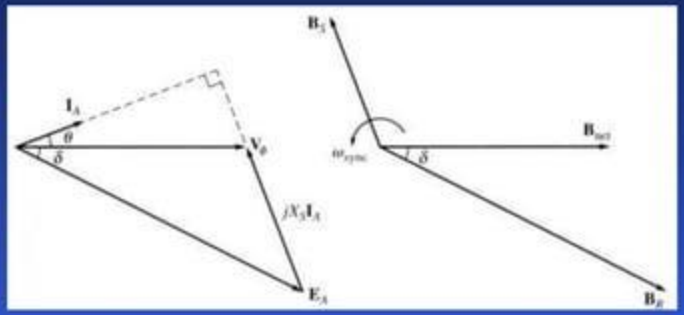
clockwise, opposing the direction of rotation. In other words, the induced torque in generators is a counter-torque that opposes the rotation caused by external torque.

If the prime mover loses power, the rotor will slow down and the rotor field B_R will fall behind the magnetic field in the machine B_{net}. Therefore, the operation of the machine changes…

Synchronous motor vs. synchronous generator

The induced torque becomes counter-clockwise, being now in the direction of rotation. The machine starts acting as a motor.

The increasing torque angle δ results in an increasing torque in the direction of rotation until it equals to the load torque.



At this point, the machine operates at steady state and synchronous speed but as a motor.

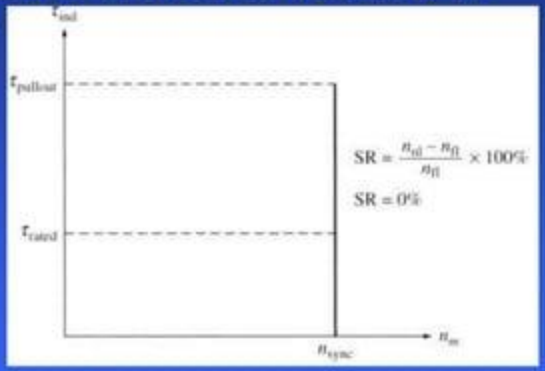
Notice that, since the direction of I_A is changed between the generator and motor actions, the polarity of stator voltage ($-jX_SI_A$) also changes.

In a summary: in a generator, E_A lies ahead of V_ϕ, while in a motor, E_A lies behind V_ϕ.

Steady-state operation of motor: Torque-speed curve

Usually, synchronous motors are connected to large power systems (infinite bus); therefore, their terminal voltage and system frequency are constant regardless the motor load. Since the motor speed is locked to the electrical frequency, the speed should be constant regardless the load.

The steady-state speed of the motor is constant from no-load to the maximum torque that motor can supply (pullout torque). Therefore, the speed regulation of synchronous motor is 0%.



The induced torque is

$$\tau_{ind} = kB_R B_{net} \sin\delta \qquad (7.72.1)$$

or

$$\tau_{ind} = \frac{3V_\phi E_A}{\omega_m X_S} \sin\delta \qquad (7.72.2)$$

Steady-state operation of motor: Torque-speed curve

The maximum pullout torque occurs when $\delta = 90^0$:

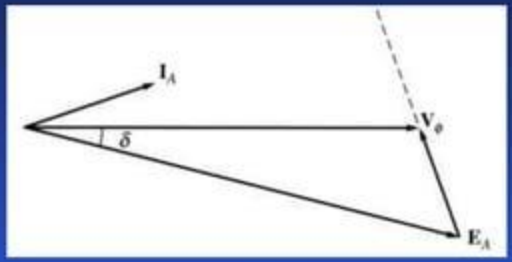
$$\tau_{\max} = kB_R B_{net} = \frac{3V_\phi E_A}{\omega_m X_S} \qquad (7.73.1)$$

Normal full-load torques are much less than that (usually, about 3 times smaller).

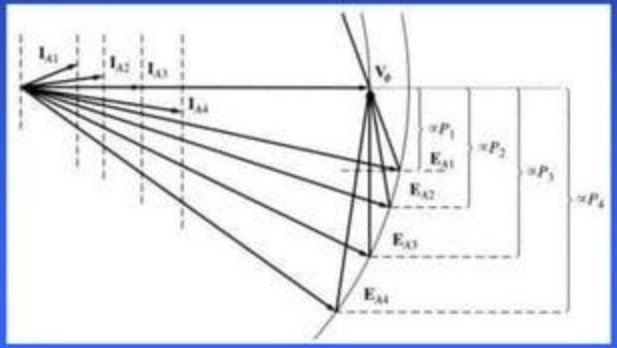
When the torque on the shaft of a synchronous motor exceeds the pullout torque, the rotor can no longer remain locked to the stator and net magnetic fields. It starts to slip behind them. As the motor slows down, the stator magnetic field "laps" it repeatedly, and the direction of the induced torque in the rotor reverses with each pass. As a result, huge torque surges of alternating direction cause the motor vibrate severely. The loss of synchronization after the pullout torque is exceeded is known as slipping poles.

Effect of torque changes

Assuming that a synchronous motor operates initially with a leading PF.
If the load on the motor increases, the rotor initially slows down increasing the torque angle δ. As a result, the induced torque increases speeding up the rotor up to the synchronous speed with a larger torque angle δ.



Since the terminal voltage and frequency supplied to the motor are constant, the magnitude of internal generated voltage must be constant at the load changes ($E_A = K\phi\omega$ and field current is constant).



Steady-state operation of motor: Effect of torque changes

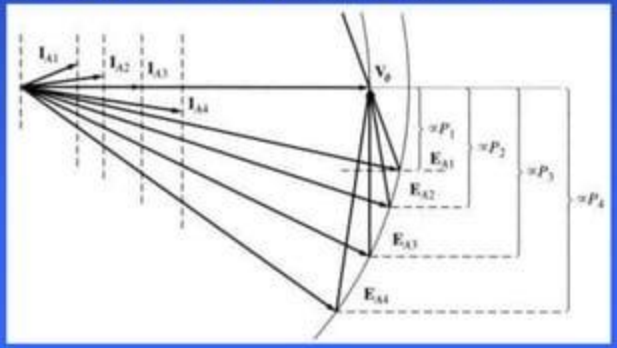
Assuming that the armature resistance is negligible, the power converted from electrical to mechanical form in the motor will be the same as its input power:

$$P = 3V_\phi I_A \cos\theta = \frac{3V_\phi E_A}{X_S}\sin\delta \qquad (7.73.1)$$

Since the phase voltage is constant, the quantities $I_A cos\theta$ and $E_A sin\delta$ are directly proportional to the power supplied by (and to) the motor. When the power supplied by the motor increases, the distance proportional to power increases.

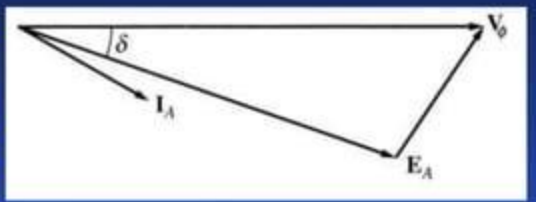
Since the internal generated voltage is constant, its phasor "swings down" as load increases. The quantity $jX_S I_A$ has to increase; therefore, the armature current I_A increases too.

Also, the PF angle changes too moving from leading to lagging.

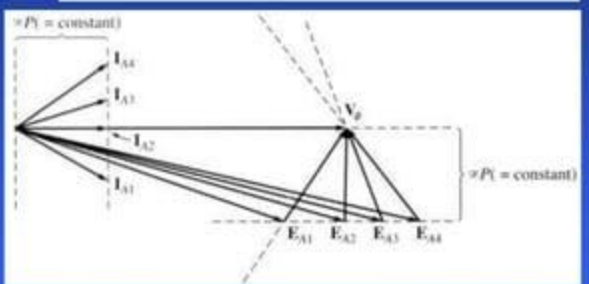


Steady-state operation of motor: Effect of field current changes

ssuming that a synchronous motor operates tially with a lagging PF.
for the constant load, the field current on the otor increases, the magnitude of the internal enerated voltage E_A increases.



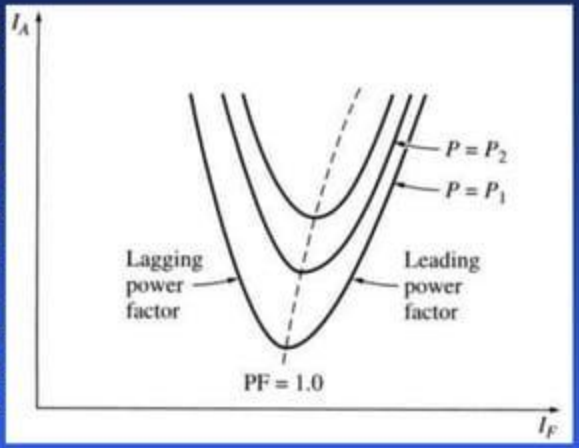
nce changes in I_A do not affect the shaft peed and the motor load is constant, the al power supplied by the motor is nchanged. Therefore, the distances oportional to power on the phasor agram ($E_A sin\delta$ and $I_A cos\theta$) must be nstant.



otice that as E_A increases, the magnitude of the armature current I_A first ecreases and then increases again. At low E_A, the armature current is lagging and e motor is an inductive load that consumes reactive power Q. As the field current creases , I_A eventually lines up with V_ϕ, and the motor is purely resistive. As the ld current further increases, I_A becomes leading and the motor is a capacitive ad that supplies reactive power Q to the system (consumes –Q).

Steady-state operation of motor: Effect of field current changes

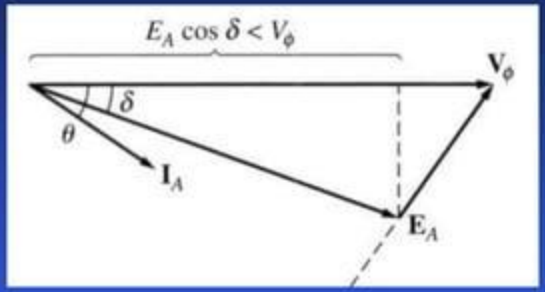
A plot of armature current vs. field current is called a synchronous motor V curve. V curves for different levels of real power have their minimum at unity PF, when only real power is supplied to the motor. For field currents less than the one giving the minimum I_A, the armature current is lagging and the motor consumes reactive power. For field currents greater than the one giving the minimum I_A, the armature current is leading and the motor supplies reactive power to the system.



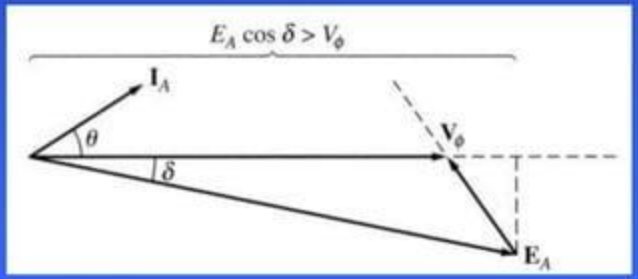
Therefore, by controlling the field current of a synchronous motor, the reactive power consumed or supplied to the power system can be controlled.

Steady-state operation of motor: Effect of field current changes

When the projection of the phasor E_A onto V_ϕ ($E_A \cos\delta$) is shorter than V_ϕ, a synchronous motor has a lagging current and consumes Q. Since the field current is small in this situation, the motor is sais to be **under-excited**.

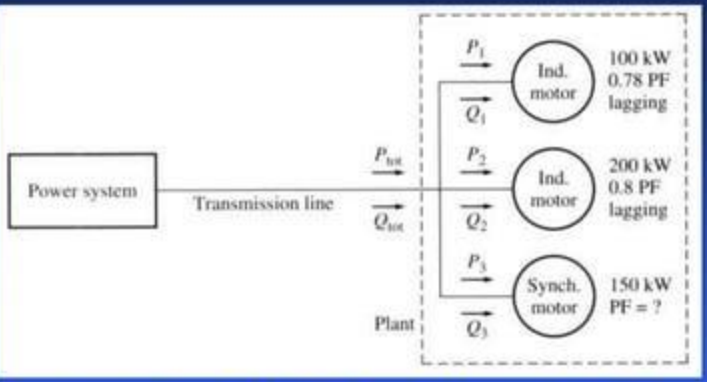


When the projection of the phasor E_A onto V_ϕ ($E_A \cos\delta$) is longer than V_ϕ, a synchronous motor has a leading current and supplies Q to the system. Since the field current is large in this situation, the motor is sais to be **over-excited**.



Steady-state operation of motor: power factor correction

Assuming that a load contains a synchronous motor (whose PF can be adjusted) in addition to motors of other types. What does the ability to set the PF of one of the loads do for the power system?



Let us consider a large power system operating at 480 V. Load 1 is an induction motor consuming 100 kW at 0.78 PF lagging, and load 2 is an induction motor consuming 200 kW at 0.8 PF lagging. Load 3 is a synchronous motor whose real power consumption is 150 kW.

a. If the synchronous motor is adjusted to 0.85 PF lagging, what is the line current?
b. If the synchronous motor is adjusted to 0.85 PF leading, what is the line current?
c. Assuming that the line losses are $P_{LL} = 3I_L^2R_L$, how du these losses compare in the two cases?

Steady-state operation of motor: power factor correction

a. The real power of load 1 is 100 kW, and the reactive power of load 1 is

$$Q_1 = P_1 \tan\theta = 100 \tan\left(\cos^{-1} 0.78\right) = 80.2\ kVAR$$

The real power of load 2 is 200 kW, and the reactive power of load 2 is

$$Q_2 = P_2 \tan\theta = 200 \tan\left(\cos^{-1} 0.8\right) = 150\ kVAR$$

The real power of load 3 is 150 kW, and the reactive power of load 3 is

$$Q_3 = P_3 \tan\theta = 150 \tan\left(\cos^{-1} 0.85\right) = 93\ kVAR$$

The total real load is $P_{tot} = P_1 + P_2 + P_3 = 100 + 200 + 150 = 450\ kW$

The total reactive load is $Q_{tot} = Q_1 + Q_2 + Q_3 = 80.2 + 150 + 93 = 323.2\ kVAR$

The equivalent system PF is

$$PF = \cos\theta = \cos\left(\tan^{-1}\frac{Q}{P}\right) = \cos\left(\tan^{-1}\frac{323.2}{450}\right) = 0.812\ \ lagging$$

The line current is

$$I_L = \frac{P_{tot}}{\sqrt{3} V_L \cos\theta} = \frac{450\,000}{\sqrt{3} \cdot 480 \cdot 0.812} = 667\ A$$

Steady-state operation of motor: power factor correction

b. The real and reactive powers of loads 1 and 2 are the same. The reactive power of load 3 is

$$Q_3 = P_3 \tan\theta = 150 \tan\left(-\cos^{-1} 0.85\right) = -93\ kVAR$$

The total real load is $P_{tot} = P_1 + P_2 + P_3 = 100 + 200 + 150 = 450\ kW$

The total reactive load is $Q_{tot} = Q_1 + Q_2 + Q_3 = 80.2 + 150 - 93 = 137.2\ kVAR$

The equivalent system PF is

$$PF = \cos\theta = \cos\left(\tan^{-1}\frac{Q}{P}\right) = \cos\left(\tan^{-1}\frac{137.2}{450}\right) = 0.957\ \ lagging$$

The line current is

$$I_L = \frac{P_{tot}}{\sqrt{3} V_L \cos\theta} = \frac{450\,000}{\sqrt{3} \cdot 480 \cdot 0.957} = 566\ A$$

Steady-state operation of motor: power factor correction

c. The transmission line losses in the first case are

$$P_{LL} = 3I_L^2 R_L = 1\,344\,700\, R_L$$

The transmission line losses in the second case are

$$P_{LL} = 3I_L^2 R_L = 961\,070\, R_L$$

We notice that the transmission power losses are 28% less in the second case, while the real power supplied to the loads is the same.

Steady-state operation of motor: power factor correction

The ability to adjust the power factor of one or more loads in a power system can significantly affect the efficiency of the power system: the lower the PF, the greater the losses in the power lines. Since most loads in a typical power system are induction motors, having one or more over-excided synchronous motors (leading loads) in the system is useful for the following reasons:

1. A leading load supplies some reactive power to lagging loads in the system. Since this reactive power does not travel along the transmission line, transmission line current is reduced reducing power losses.
2. Since the transmission line carries less current, the line can be smaller for a given power flow reducing system cost.
3. The over-excited mode of synchronous motor increases the motor's maximum torque.

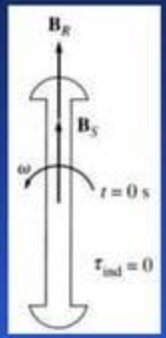
Usage of synchronous motors or other equipment increasing the overall system's PF is called power-factor correction. Since a synchronous motor can provide PF correction, many loads that can accept constant speed are driven by over-excited synchronous motors.

Starting synchronous motors

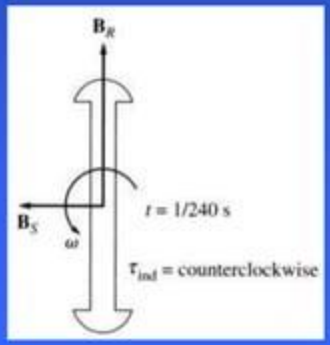
Consider a 60 Hz synchronous motor.
When the power is applied to the stator windings, the rotor (and, therefore its magnetic field B_R) is stationary. The stator magnetic field B_S starts sweeping around the motor at synchronous speed.
Note that the induced torque on the shaft

$$\tau_{ind} = kB_R \times B_S \qquad (7.84.1)$$

is zero at t = 0 since both magnetic fields are aligned.

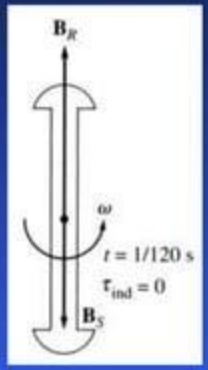


At t = 1/240 s the rotor has barely moved but the stator magnetic field B_S has rotated by 90^0. Therefore, the torque on the shaft is non-zero and counter-clockwise.

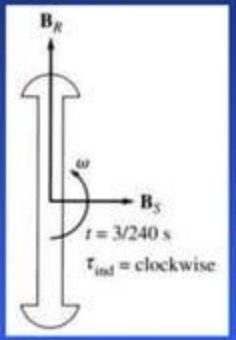


Starting synchronous motors

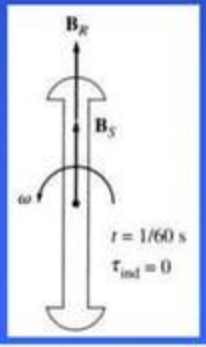
At t = 1/120 s the rotor and stator magnetic fields point in opposite directions, and the induced torque on the shaft is zero again.



At t = 3/240 s the stator magnetic fields point to the right, and the induced torque on the shaft is non-zero but clockwise.



Finally, at t = 1/60 s the rotor and stator magnetic fields are aligned again, and the induced torque on the shaft is zero.



During one electrical cycle, the torque was counter-clockwise and then clockwise, and the average torque is zero. The motor will vibrate heavily and finally overheats!

Starting synchronous motors

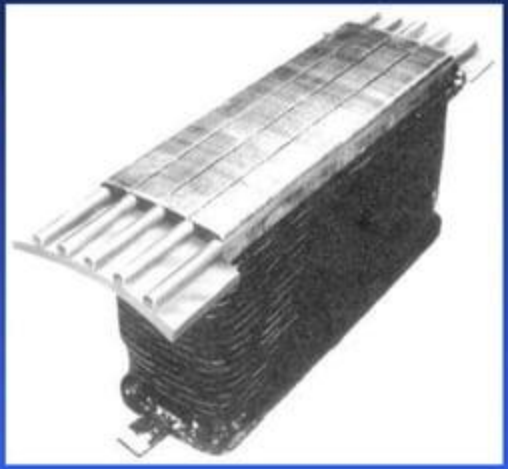
Three basic approaches can be used to safely start a synchronous motor:

1. Reduce the speed of the stator magnetic field to a low enough value that the rotor can accelerate and two magnetic fields lock in during one half-cycle of field rotation. This can be achieved by reducing the frequency of the applied electric power (which used to be difficult but can be done now).
2. Use an external prime mover to accelerate the synchronous motor up to synchronous speed, go through the paralleling procedure, and bring the machine on the line as a generator. Next, turning off the prime mover will make the synchronous machine a motor.
3. Use damper windings or amortisseur windings – the most popular.

Motor starting by amortisseur or damper windings

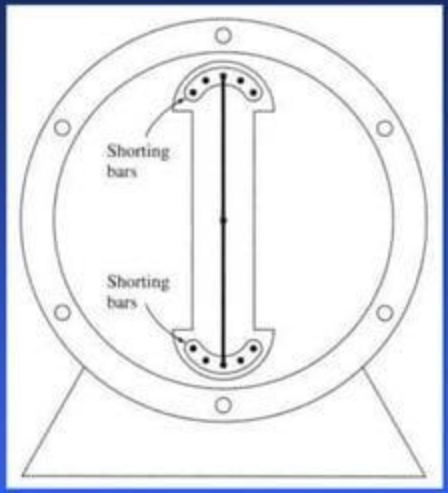
Amortisseur (damper) windings are special bars laid into notches carved in the rotor face and then shorted out on each end by a large shorting ring.

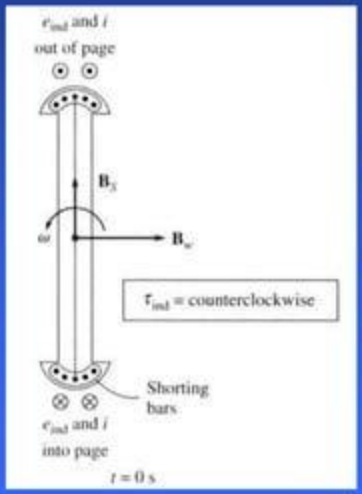




Motor starting by amortisseur or damper windings

A diagram of a salient 2-pole rotor with an amortisseur winding, with the shorting bars on the ends of the two rotor pole faces connected by wires (not quite the design of actual machines).





We assume initially that the rotor windings are disconnected and only a 3-phase set of voltages are applied to the stator.

At $t = 0$, assume that B_S (stator field) is vertical.

As B_S sweeps along in s counter-clockwise direction, it induces a voltage in bars of the amortisseur winding:

$$e_{ind} = (v \times B) \cdot l \qquad (7.88.1)$$

Motor starting by amortisseur or damper windings

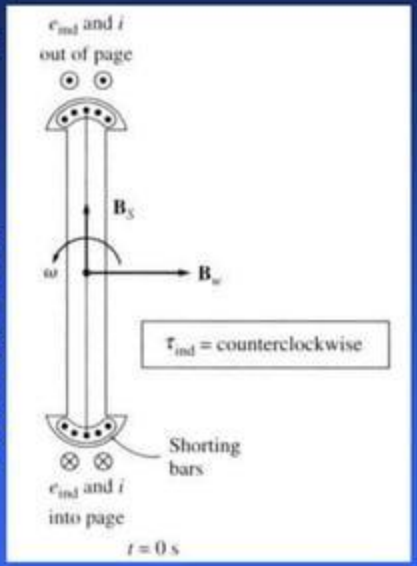
Here v – the velocity of the bar relative to the magnetic field;

B – magnetic flux density vector;

l – length of conductor in the magnetic field.

The bars at the top of the rotor are moving to the right relative to the magnetic field: a voltage, with direction out of page, will be induced. Similarly, the induced voltage is into the page in the bottom bars. These voltages produce a current flow out of the top bars and into the bottom bars generating a winding magnetic field B_w to the right. Two magnetic fields will create a torque

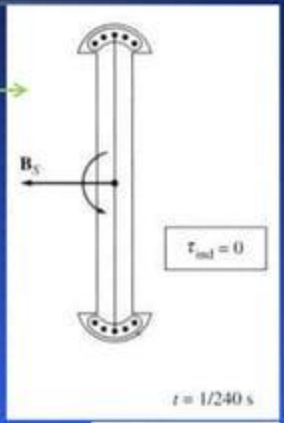
$$\tau_{ind} = kB_W \times B_S \qquad (7.89.1)$$

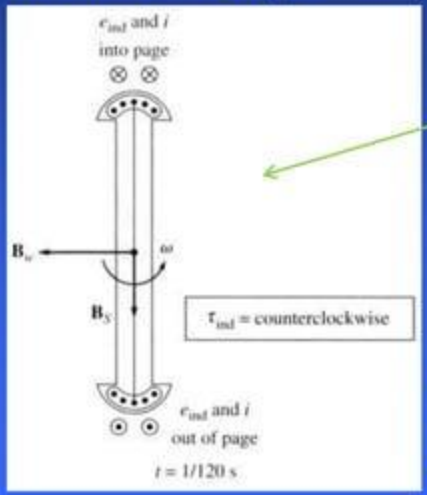


The resulting induced torque will be counter-clockwise.

Motor starting by amortisseur or damper windings

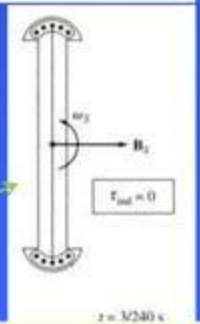
At t = 1/240 s, B_S has rotated 90^0 while the rotor has barely moved. Since v is parallel to B_S, the voltage induced in the amortisseur windings is zero, therefore, no current in wires create a zero-torque.





At t = 1/120 s, B_S has rotated another 90^0 and the rotor is still. The voltages induced in the bars create a current inducing a magnetic field pointing to the left. The torque is counter-clockwise.

Finally, at t = 3/240 s, no voltage is induced in the amortisseur windings and, therefore, the torque will be zero.



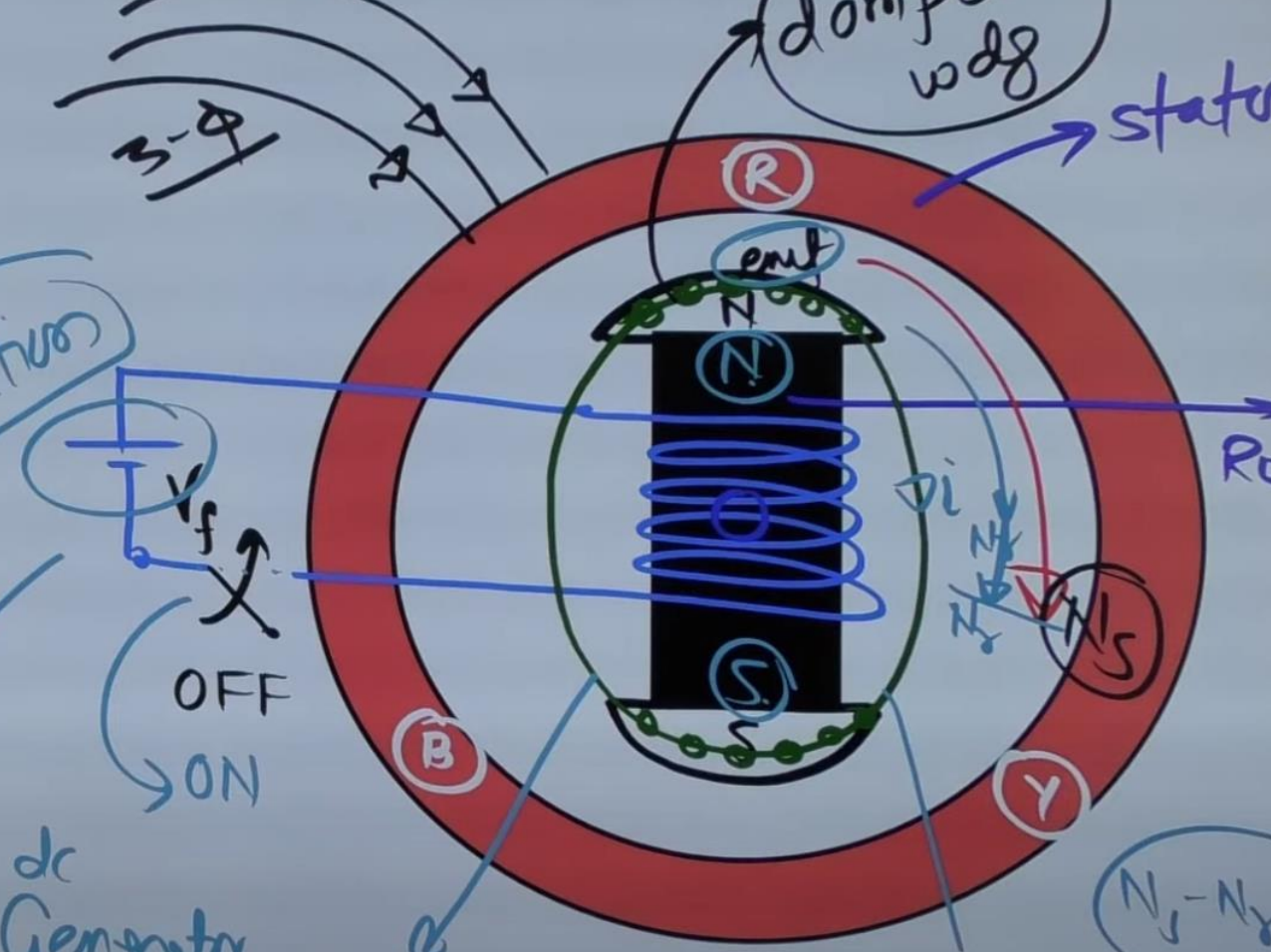
Motor starting by amortisseur or damper windings

We observe that the torque is either counter-clockwise or zero, but it is always unidirectional. Since the net torque is nonzero, the motor will speed up.

However, the rotor will never reach the synchronous speed! If a rotor was running at the synchronous speed, the speed of stator magnetic field B_S would be the same as the speed of the rotor and, therefore, no relative motion between the rotor and the stator magnetic field. If there is no relative motion, no voltage is induced and, therefore, the torque will be zero.

Instead, when the rotor's speed is close to synchronous, the regular field current can be turned on and the motor will operate normally. In real machines, field circuit are shorted during starting. Therefore, if a machine has damper winding:

1. Disconnect the field windings from their DC power source and short them out;
2. Apply a 3-phase voltage to the stator and let the rotor to accelerate up to near-synchronous speed. The motor should have no load on its shaft to enable motor speed to approach the synchronous speed as closely as possible;
3. Connect the DC field circuit to its power source: the motor will lock at synchronous speed and loads may be added to the shaft.

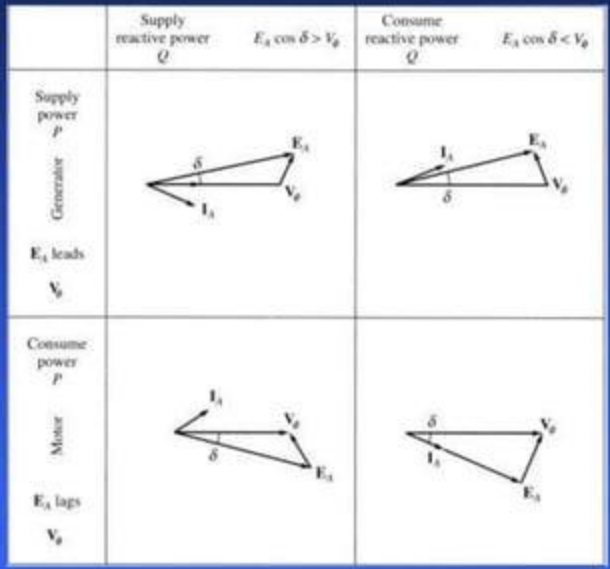

3-ϕ
R
N
S
B
Y
OFF
ON
dc

92

Relationship between synchronous generators and motors

Synchronous generator and synchronous motor are physically the same machines!
A synchronous machine can supply real power to (generator) or consume real power (motor) from a power system. It can also either consume or supply reactive power to the system.

1. The distinguishing characteristic of a synchronous generator (supplying P) is that E_A lies ahead of V_ϕ while for a motor E_A lies behind V_ϕ.
2. The distinguishing characteristic of a machine supplying reactive power Q is that $E_a cos\delta > V_\phi$ (regardless whether it is a motor or generator). The machine consuming reactive power Q has $E_a cos\delta < V_\phi$.



Synchronous machine ratings

The speed and power that can be obtained from a synchronous motor or generator are limited. These limited values are called ratings of the machine. The purpose of ratings is to protect the machine from damage. Typical ratings of synchronous machines are voltage, speed, apparent power (kVA), power factor, field current and service factor.

1. Voltage, Speed, and Frequency

The rated frequency of a synchronous machine depends on the power system to which it is connected. The commonly used frequencies are 50 Hz (Europe, Asia), 60 Hz (Americas), and 400 Hz (special applications: aircraft, spacecraft, etc.). Once the operation frequency is determined, only one rotational speed in possible for the given number of poles:

$$n_m = \frac{120 f_e}{P} \qquad (7.93.1)$$

Synchronous machine ratings

A generator's voltage depends on the flux, the rotational speed, and the mechanical construction of the machine. For a given design and speed, the higher the desired voltage, the higher the flux should be. However, the flux is limited by the field current.

The rated voltage is also limited by the windings insulation breakdown limit, which should not be approached closely.

Is it possible to operate a synchronous machine at a frequency other than the machine is rated for? For instance, can a 60 Hz generator operate at 50 Hz?

The change in frequency would change the speed. Since $E_A = K\phi\omega$, the maximum allowed armature voltage changes when frequency changes. Specifically, if a 60 Hz generator will be operating at 50 Hz, its operating voltage must be derated to 50/60 or 83.3 %.

Synchronous machine ratings

2. Apparent power and Power factor

Two factors limiting the power of electric machines are
1) Mechanical torque on its shaft (usually, shaft can handle much more torque)
2) Heating of the machine's winding

The practical steady-state limits are set by heating in the windings.
The maximum acceptable armature current sets the apparent power rating for a generator:

$$S = 3V_{\phi} I_A \tag{7.95.1}$$

If the rated voltage is known, the maximum accepted armature current determines the apparent power rating of the generator:

$$S = 3V_{\phi,rated} I_{A,\max} = \sqrt{3} V_{L,rated} I_{L,\max} \tag{7.95.2}$$

The power factor of the armature current is irrelevant for heating the armature windings.

Synchronous machine ratings

The stator cupper losses also do not depend on the current angle:

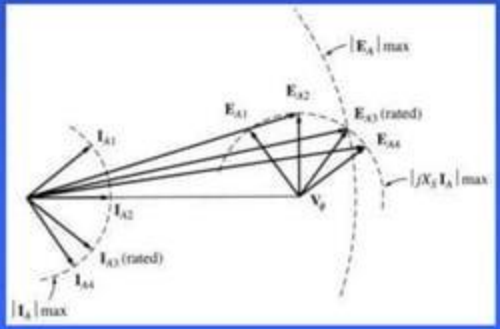
$$P_{SCL} = 3I_A^2 R_A \qquad (7.96.1)$$

Since the current angle is irrelevant to the armature heating, synchronous generators are rated in kVA rather than in KW.

The rotor (field winding) cupper losses are:

$$P_{RCL} = I_F^2 R_F \qquad (7.96.2)$$

Allowable heating sets the maximum field current, which determines the maximum acceptable armature voltage E_A. These translate to restrictions on the lowest acceptable power factor: The current I_A can have different angles (that depends on PF). E_A is a sum of V_ϕ and $jX_S I_A$. We see that, (for a constant V_ϕ) for some angles the required E_A exceeds its maximum value.



Synchronous machine ratings

If the armature voltage exceeds its maximum allowed value, the windings could be damaged. The angle of I_A that requires maximum possible E_A specifies the rated power factor of the generator. It is possible to operate the generator at a lower (more lagging) PF than the rated value, but **only** by decreasing the apparent power supplied by the generator.

Synchronous motors are usually rated in terms of real output power and the lowest PF at full-load conditions.

3. Short-time operation and service factor

A typical synchronous machine is often able to supply up to 300% of its rated power for a while (until its windings burn up). This ability to supply power above the rated values is used to supply momentary power surges during motor starts.

It is also possible to use synchronous machine at powers exceeding the rated values for longer periods of time, as long as windings do not have time to hit up too much before the excess load is removed. For instance, a generator that could supply 1 MW indefinitely, would be able to supply 1.5 MW for 1 minute without serious harm and for longer periods at lower power levels.

Synchronous Machine ratings

The maximum temperature rise that a machine can stand depends on the insulation class of its windings. The four standard insulation classes with they temperature ratings are:

A – 60^0C above the ambient temperature
B – 80^0C above the ambient temperature
F – 105^0C above the ambient temperature
H – 125^0C above the ambient temperature

The higher the insulation class of a given machine, the greater the power that can be drawn out of it without overheating its windings.

The overheating is a serious problem and synchronous machines should not be overheated unless absolutely necessary. However, power requirements of the machine not always known exactly prior its installation. Because of this, general-purpose machines usually have their service factor defined as the ratio of the actual maximum power of the machine to the rating on its plate.
For instance, a machine with a service factor of 1.15 can actually be operated at 115% of the rated load indefinitely without harm.